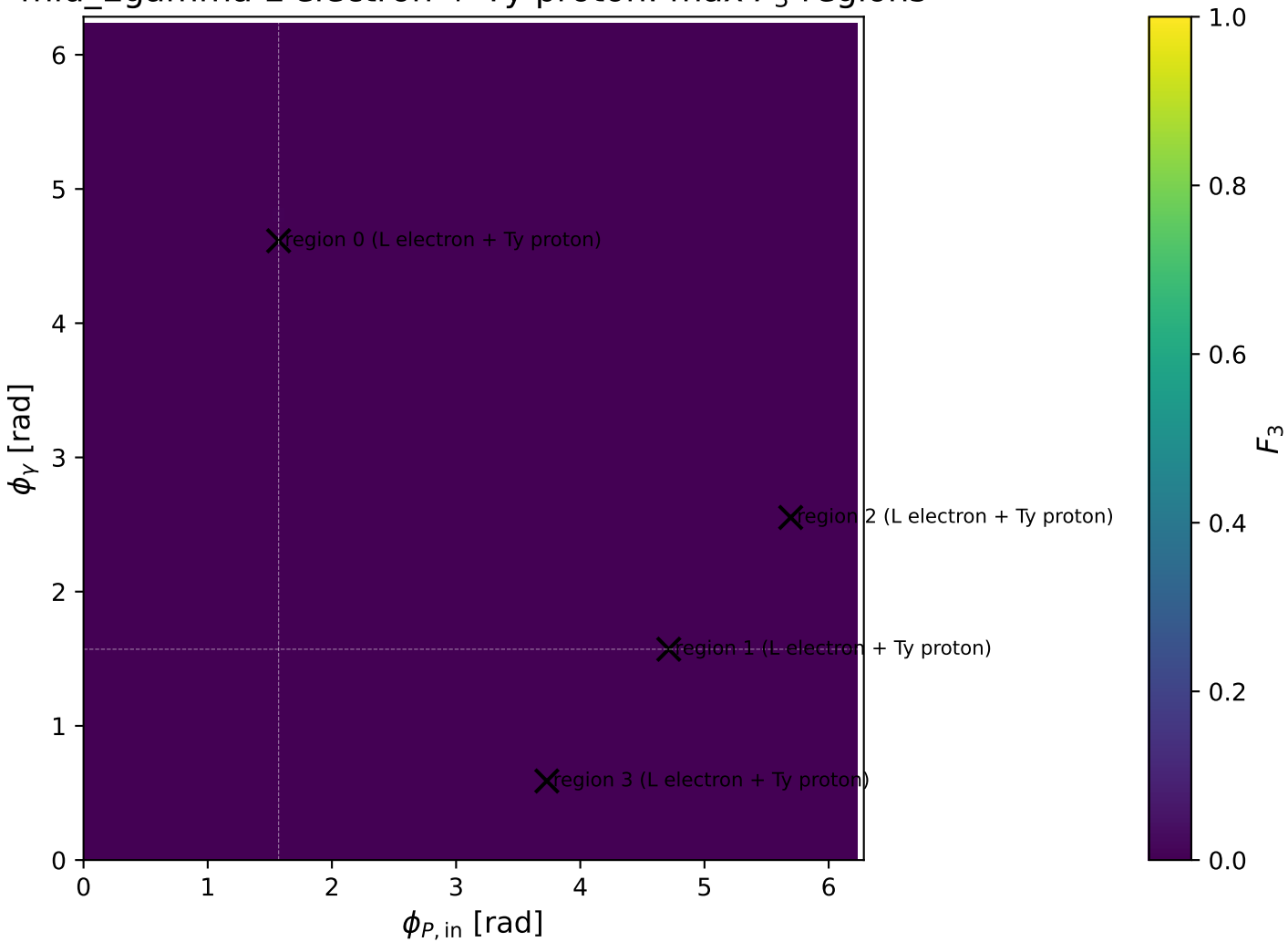
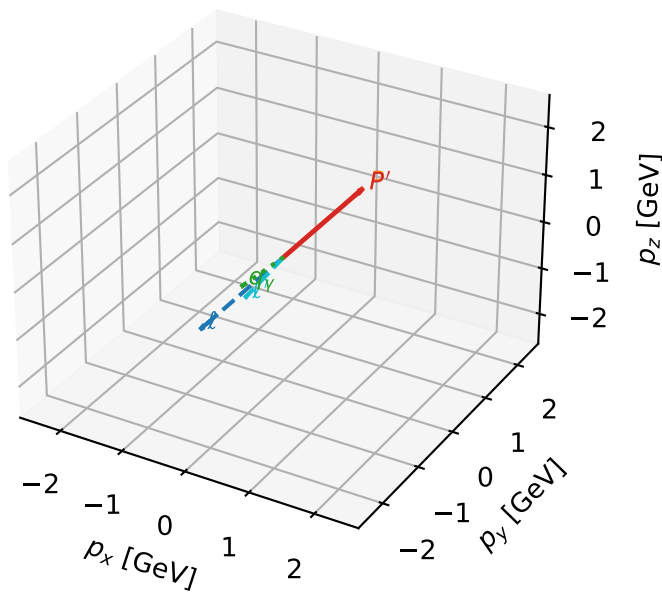


mid\_Egamma L electron + Ty proton: max  $F_3$  regions



# L electron + Ty proton characteristic max $F_3$ configuration

3D momenta

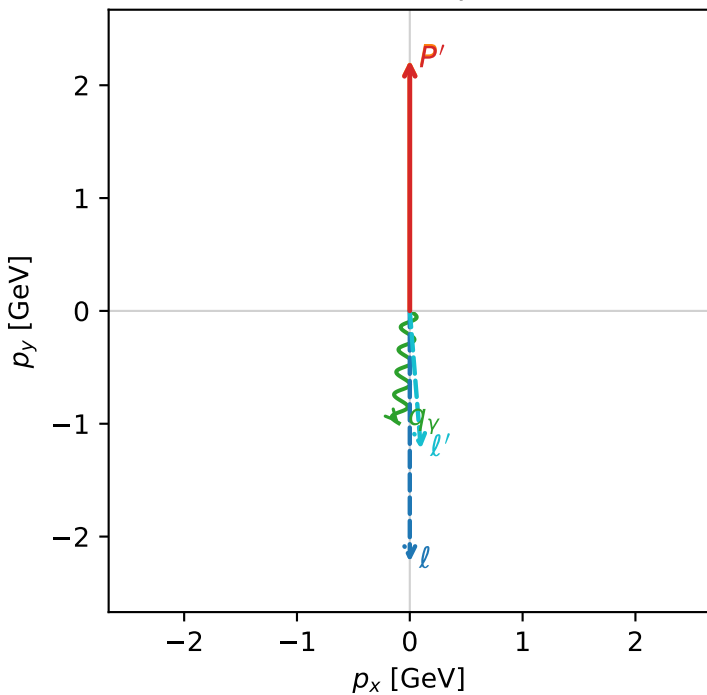


f3\_mid\_egamma\_lty\_region\_0 F\_3=0.00391829  
 region: mid\_Egamma\_LTy\_region\_0  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_mid\_Egamma  
 incoming state:  $|L_e\rangle \otimes |T_{y,p}\rangle$

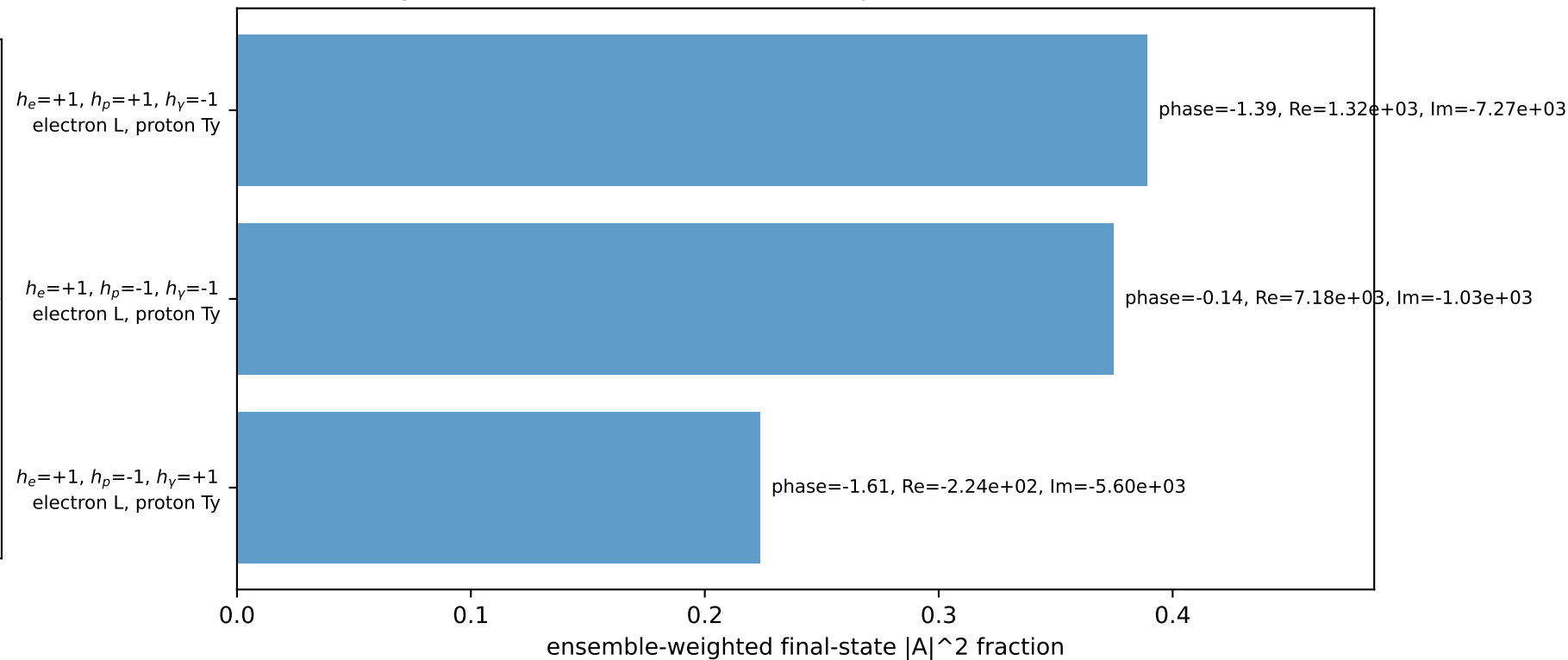
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.21987$   
 $\theta_{in}=1.57079$ ,  $\phi_l=4.71239$ ,  $\phi_p=1.5708$   
 $E_\gamma=1$ ,  $\phi_\gamma=4.61421$   
 $Q^2=0.0174222$ ,  $x_B=0.00188234$ ,  $t=-3.12347e-06$ ,  $W^2=10.118$ ,  $y=0.448518$   
 $\theta(l', \gamma)=10.2009$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $l'$   $E= 1.2286$   $p_x= 0.0980$   $p_y= -1.2247$   $p_z= 0.0000$   
 $P'$   $E= 2.4099$   $p_x= 0.0000$   $p_y= 2.2199$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.0000$   $p_x= -0.0980$   $p_y= -0.9952$   $p_z= 0.0000$

Transverse plane

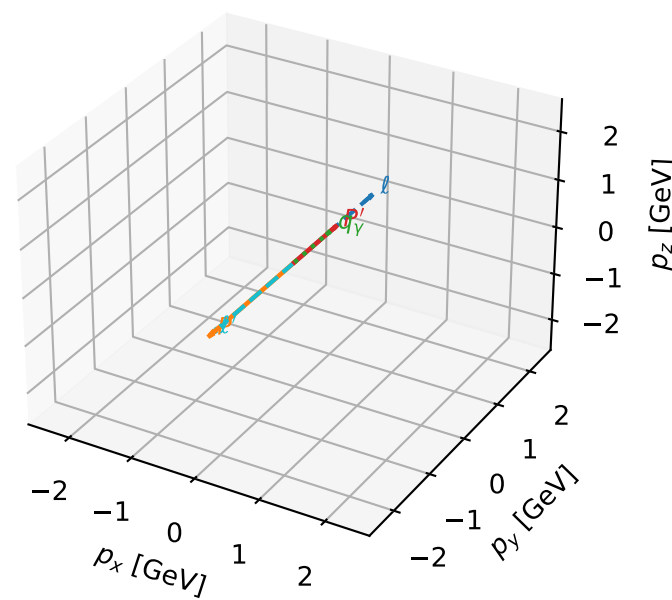


Leading initial-ensemble/final-state components (N=3, retained=98.7%)



L electron + Ty proton characteristic max  $F_3$  configuration

3D momenta

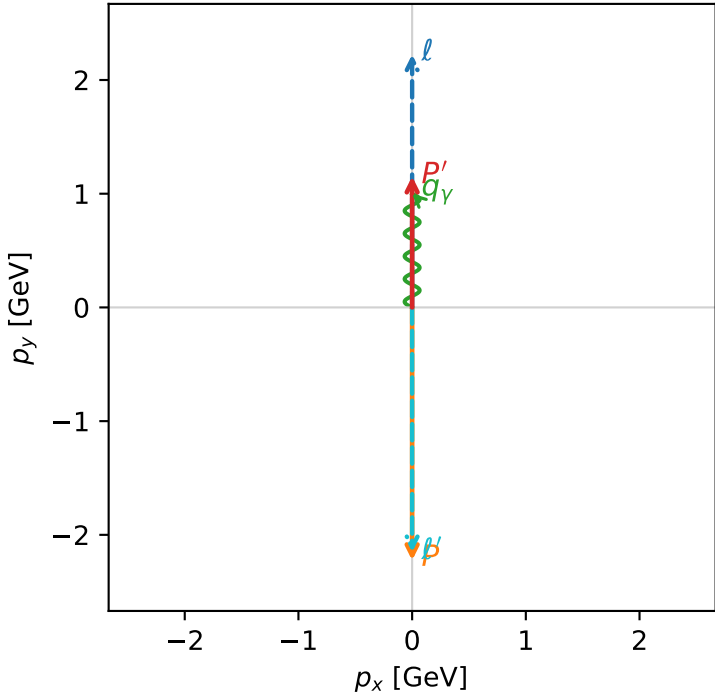


f3\_mid\_egamma\_lty\_region\_1 F\_3=0.000464679  
region: mid\_Egamma\_LTy\_region\_1  
outgoing-state purity: 1  
kinematic point: high\_s\_high\_theta\_in\_mid\_Egamma  
incoming state:  $|L_e\rangle \otimes |\bar{T}_{\gamma,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=1.15253$   
 $\theta_{in}=1.57079$ ,  $\phi_t=1.5708$ ,  $\phi_P=4.71239$   
 $E_\gamma=1$ ,  $\phi_\gamma=1.5708$   
 $Q^2=19.1525$ ,  $x_B=0.96635$ ,  $t=-10.5424$ ,  $W^2=1.54677$ ,  $y=0.960429$   
 $\theta(\ell',\gamma)=180$  deg

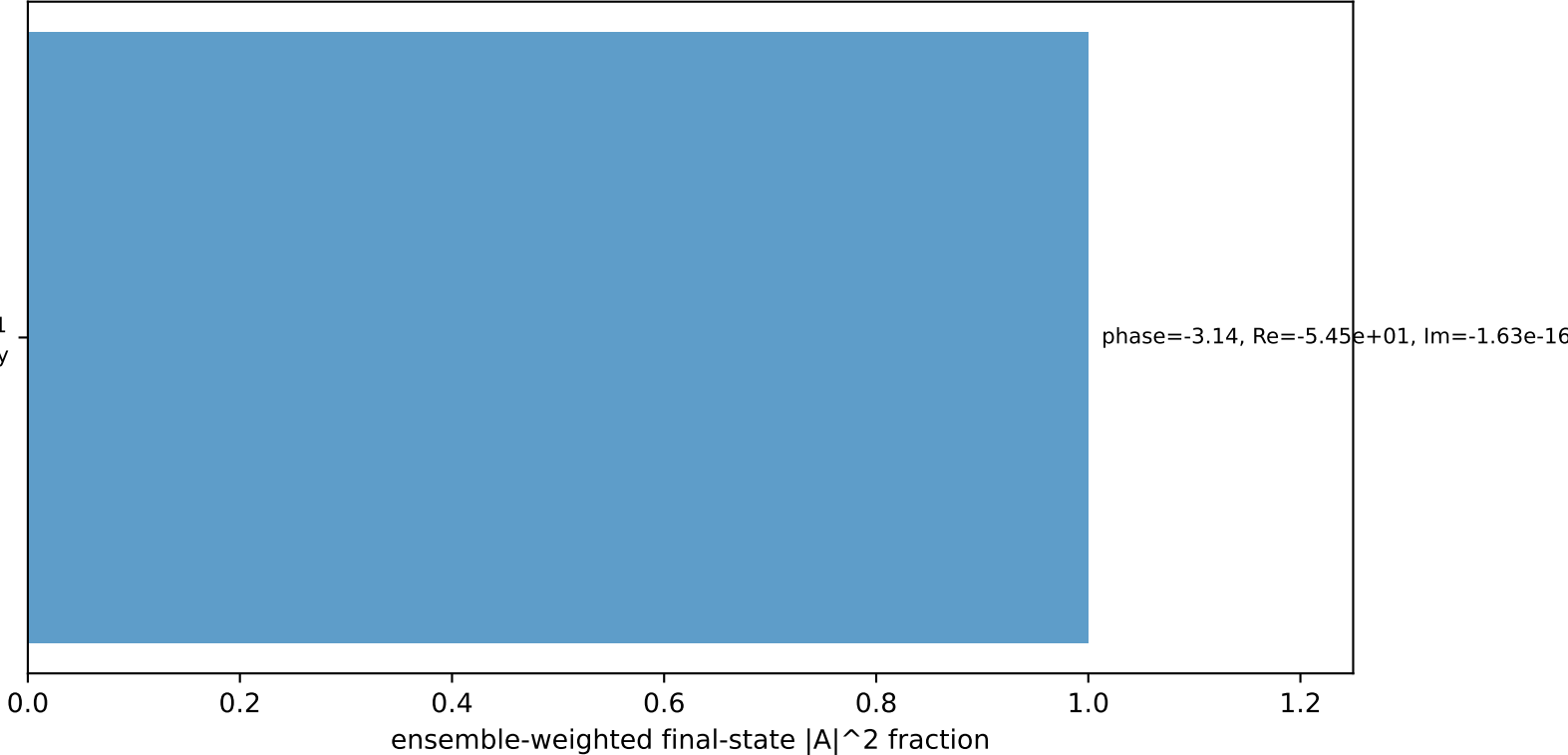
four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= 0.0000$   
 $\ell'$   $E= 2.1525$   $p_x= -0.0000$   $p_y= -2.1525$   $p_z= 0.0000$   
 $P'$   $E= 1.4860$   $p_x= 0.0000$   $p_y= 1.1525$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.0000$   $p_x= 0.0000$   $p_y= 1.0000$   $p_z= 0.0000$

Transverse plane



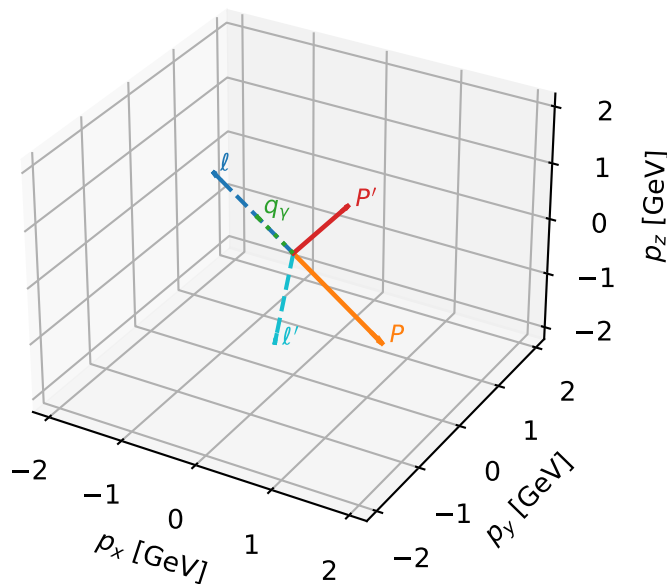
$h_e=-1$ ,  $h_p=-1$ ,  $h_\gamma=+1$   
electron L, proton Ty

Leading initial-ensemble/final-state components (N=1, retained=100.0%)



# L electron + Ty proton characteristic max $F_3$ configuration

3D momenta



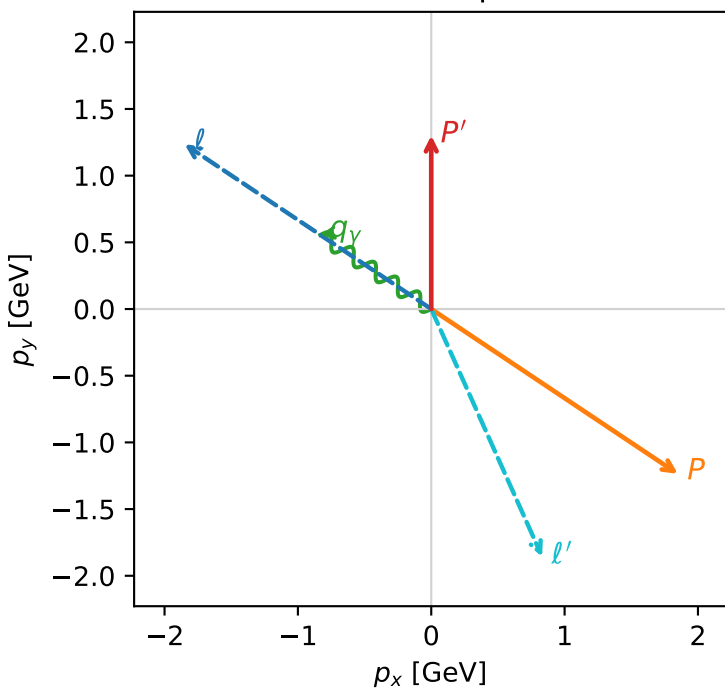
f3\_mid\_egamma\_lty\_region\_2 F\_3=0.000343043  
 region: mid\_Egamma\_LTy\_region\_2  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_mid\_Egamma  
 incoming state:  $|L_e\rangle \otimes |T_{y,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=1.30122$   
 $\theta_{in}=1.57079$ ,  $\phi_l=2.55254$ ,  $\phi_p=5.69414$   
 $E_Y=1$ ,  $\phi_Y=2.55254$   
 $Q^2=16.7159$ ,  $x_B=0.904629$ ,  $t=-9.20119$ ,  $W^2=2.64213$ ,  $y=0.895436$   
 $\theta(l', \gamma)=147.873$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:

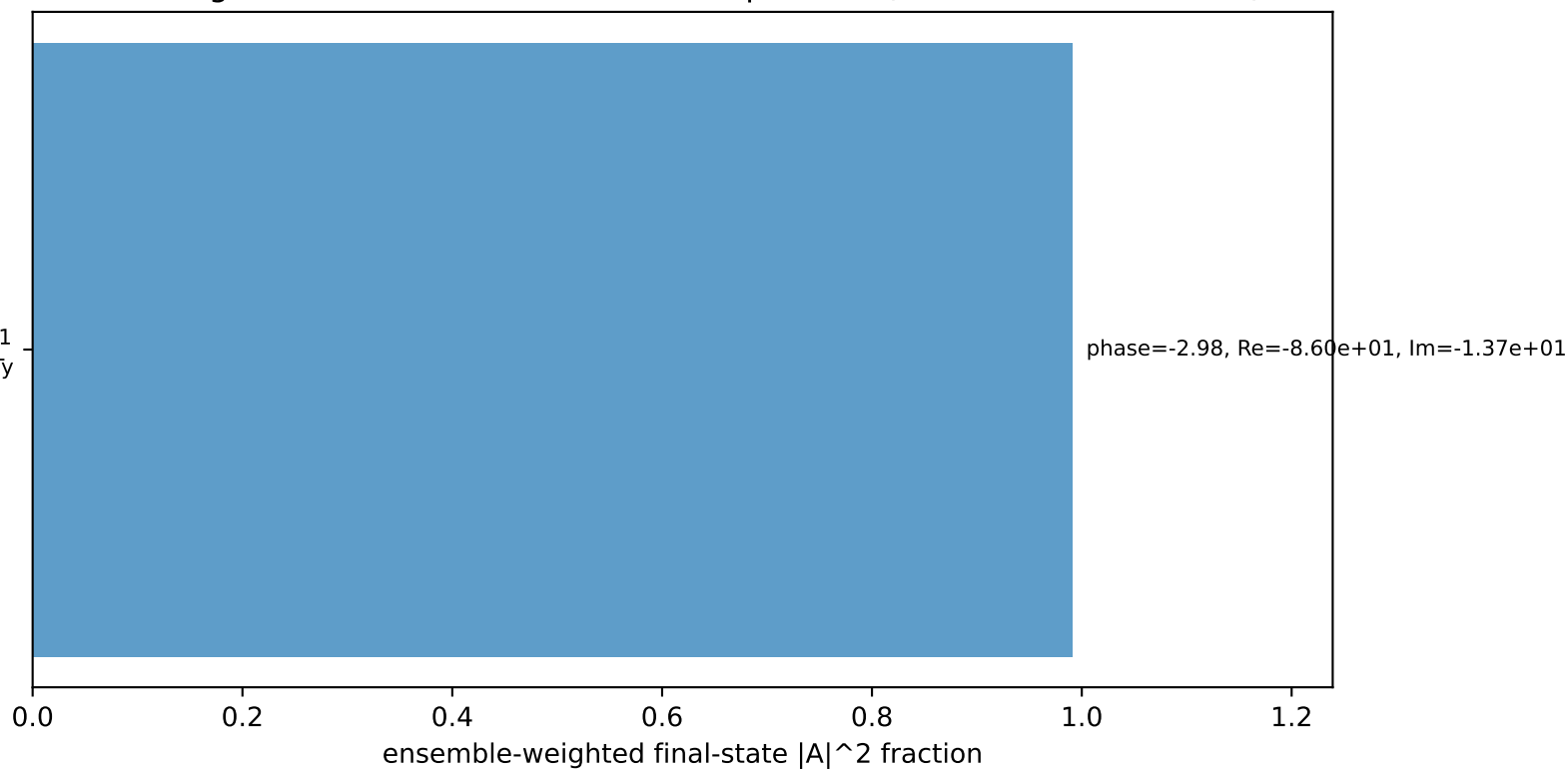
|       |      |        |        |         |        |         |        |         |
|-------|------|--------|--------|---------|--------|---------|--------|---------|
| $l$   | $E=$ | 2.2244 | $p_x=$ | -1.8495 | $p_y=$ | 1.2358  | $p_z=$ | -0.0000 |
| $P$   | $E=$ | 2.4141 | $p_x=$ | 1.8495  | $p_y=$ | -1.2358 | $p_z=$ | 0.0000  |
| $l'$  | $E=$ | 2.0345 | $p_x=$ | 0.8315  | $p_y=$ | -1.8568 | $p_z=$ | 0.0000  |
| $P'$  | $E=$ | 1.6041 | $p_x=$ | 0.0000  | $p_y=$ | 1.3012  | $p_z=$ | 0.0000  |
| $q_Y$ | $E=$ | 1.0000 | $p_x=$ | -0.8315 | $p_y=$ | 0.5556  | $p_z=$ | 0.0000  |

Transverse plane



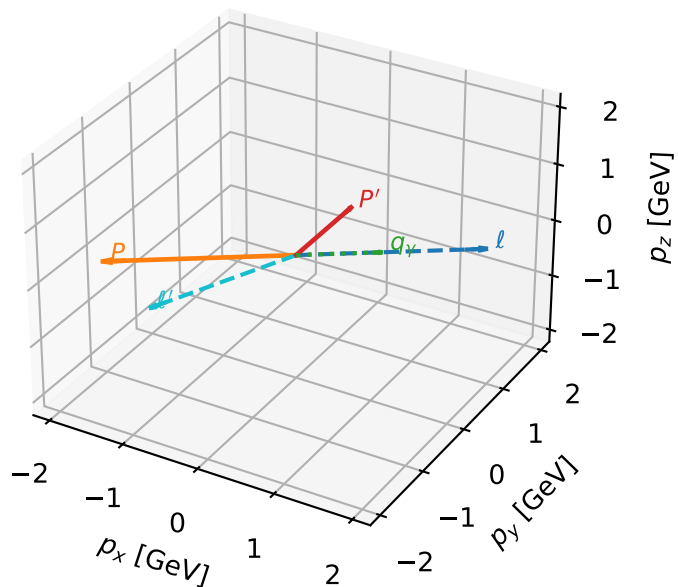
$h_e=-1$ ,  $h_p=-1$ ,  $h_Y=+1$   
 electron L, proton Ty

Leading initial-ensemble/final-state components (N=1, retained=99.1%)



# L electron + Ty proton characteristic max $F_3$ configuration

3D momenta



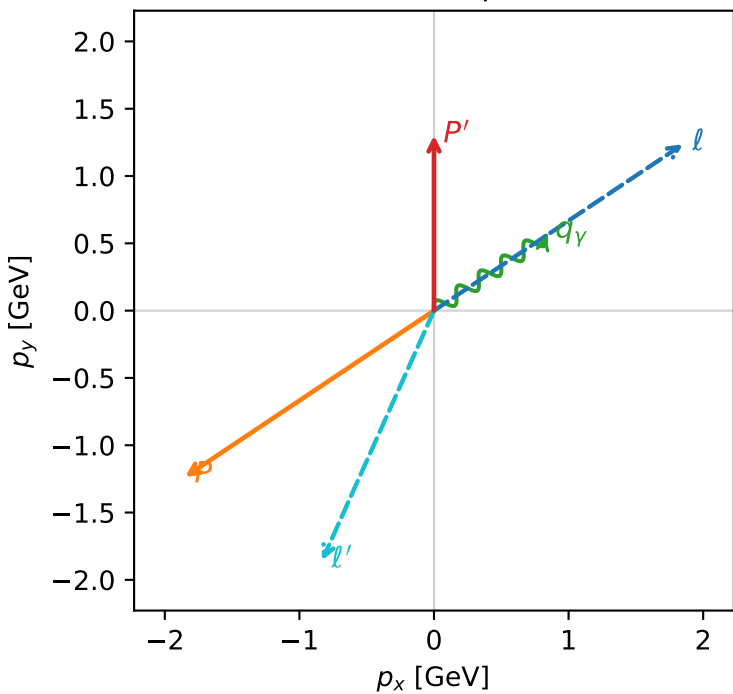
f3\_mid\_egamma\_lty\_region\_3  $F_3=0.000343043$   
 region: mid\_Egamma\_LTy\_region\_3  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_mid\_Egamma  
 incoming state:  $|L_e\rangle \otimes |T_{y,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=1.30122$   
 $\theta_{in}=1.57079$ ,  $\phi_l=0.589049$ ,  $\phi_P=3.73064$   
 $E_\gamma=1$ ,  $\phi_\gamma=0.589049$   
 $Q^2=16.7159$ ,  $x_B=0.904629$ ,  $t=-9.20119$ ,  $W^2=2.64213$ ,  $y=0.895436$   
 $\theta(l', \gamma)=147.873$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:

|            |             |                |                |                |
|------------|-------------|----------------|----------------|----------------|
| $l$        | $E=$ 2.2244 | $p_x=$ 1.8495  | $p_y=$ 1.2358  | $p_z=$ -0.0000 |
| $P$        | $E=$ 2.4141 | $p_x=$ -1.8495 | $p_y=$ -1.2358 | $p_z=$ 0.0000  |
| $l'$       | $E=$ 2.0345 | $p_x=$ -0.8315 | $p_y=$ -1.8568 | $p_z=$ 0.0000  |
| $P'$       | $E=$ 1.6041 | $p_x=$ 0.0000  | $p_y=$ 1.3012  | $p_z=$ 0.0000  |
| $q_\gamma$ | $E=$ 1.0000 | $p_x=$ 0.8315  | $p_y=$ 0.5556  | $p_z=$ 0.0000  |

Transverse plane



$h_e=-1$ ,  $h_p=-1$ ,  $h_\gamma=+1$   
 electron L, proton Ty

Leading initial-ensemble/final-state components (N=1, retained=99.1%)

